

Abstract

We study dexterous robotic piano playing in PianoMime, where policies control anthropomorphic hands to play target key-press trajectories. Our goal is to reproduce the baseline behavior and test whether targeted improvements raise note-level F1 for both a difficult single-song specialist policy and a generalist policy on unseen clips. We report successful gains together with diagnostic attempts, comparison figures, summary tables, and performance videos.

Specialist F1
0.8725 -> 0.9187
Happy_8 baseline to improved

Generalist F1
0.6150 -> 0.6256
mean over 5 test clips

Task and Motivation

Task overview

- 88-key spring keyboard in ROBOPIANIST / PianoMime.
- Two Shadow Hands execute a target key-press trajectory.
- Evaluate specialist and generalist piano-playing policies.

Why it matters

- High-DOF piano control requires timing, contact accuracy, and smooth motion.
- Plausible motion can still miss note onsets or suppress fast corrections.
- Over-smoothing or weak goal conditioning can reduce note accuracy, so we study both reward shaping and generalist-policy representations.

Problem Formulation in RL

Each clip is an episodic RL control problem. The policy observes robot state and target notes, outputs hand actions or residual corrections, and is evaluated by note-level precision, recall, and F1.

MDP $\mathcal{M} = (S, \mathcal{A}, P, r, \gamma)$
Policy $\pi_{\theta}(a_t | s_t), \quad a_t = a_t^K + \alpha r_t$
Metric $F_1 = \frac{2PR}{P+R}$

Task: imitate target key-press trajectories, then evaluate note F1.

State
proprioception, key states, future note targets

Action
hand control or residual correction

Reward
press success, mimic tracking, timing, smoothness

Metric
Precision, Recall, F1, performance video

Goal: maximize note F1 while preserving playable hand motion.

Method: Key Improvements

Core idea Specialist policy: improve note onset timing while keeping residual motion controlled. Generalist policy: improve low-level diffusion objective, goal conditioning, RP1M data alignment, and teacher-regularized recovery.

Specialist: Combined Method

Happy_8 uses one combined improvement, not separate ablations. The baseline can miss short onsets or press slightly late when residual corrections are too weak; it can also become unstable when residuals are too large. We therefore tune reward timing, residual regularization, and execution scale together.

Onset-aware reward

add extra reward on frames where a note first needs to be pressed, so the policy pays more attention to contact timing instead of only sustained key states; mainly improves recall.

Light smoothness

add small residual magnitude, residual delta, action smoothness, qvel, and qacc penalties. The weights are kept small to reduce finger jitter without suppressing fast note transitions.

Residual calibration

sweep execution residual scale after training. This avoids over-correction that causes false key presses and under-correction that leaves missed notes; best scales are around 0.0205 or 0.03.

Combined reward and residual execution

$$R = R_{\text{task}} + \lambda_o R_{\text{onset}} - \lambda_s \|\Delta a_t\|^2 - \lambda_r \|r_t\|^2$$
$$a_t = a_t^K + \alpha r_t, \quad \alpha \approx 0.0205 \text{ or } 0.03$$

Generalist: Diffusion Policy

PianoMime generalist uses a two-stage diffusion policy: high-level predicts future fingertip trajectories, then low-level predicts 46-D hand actions from trajectory and proprioception.

Smooth diffusion objective

Fine-tune the low-level diffusion model, not the high-level policy or rollout post-processing. Reconstruct the clean action estimate from the predicted noise, then supervise its local motion shape.

Denoised action estimate

$$\hat{a}_0 = (a_{\text{noisy}} - \sqrt{1 - \alpha_t} \varepsilon_{\theta}) / \sqrt{\alpha_t}$$

Training objective

$$L = L_{\varepsilon} + .01L_{\Delta} + .005L_J + .001L_{||} + .001L_{x0}$$

Trajectory terms

Match first-order action changes and second-order jerk against the demonstration. These terms are used only for low-noise DDPM timesteps $t \leq 30$, where the reconstructed clean action is reliable.

Delta matching

$$L_{\Delta} = \|\Delta \hat{a}_0 - \Delta a_{\text{demo}}\|^2$$

Jerk matching

$$L_J = \|\Delta^2 \hat{a}_0 - \Delta^2 a_{\text{demo}}\|^2$$

Goal side-channel representation

Add explicit future-goal conditioning so similar robot states can map to different actions under different upcoming note patterns.

Low-level conditioning

$$\pi_L(a_t | s_t, \tau_{t:t+4}^{HL}, g_{t:t+4})$$

Data Integration and Diagnostics

RP1M low-level alignment

Direct RP1M conversion is unreliable because its action/state schema does not match PianoMime low-level observations. The useful variant aligns fingertip ordering and learns a linear action46-to-q_hand54 mapping before low-level fine-tuning, so the imported data has a compatible hand-state representation.

$$q_{\text{hand}} = W a_{46} + b$$

RP1M recovery + teacher regularization

After RP1M low-level fine-tuning, recover on original PianoMime data while regularizing against the frozen official teacher. This tests whether broader state-action coverage can be retained without drifting away from the original action distribution.

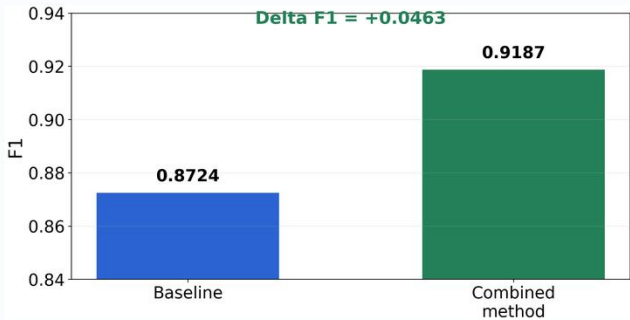
$$L = L_{\text{noise}} + \lambda \|\varepsilon_s - \varepsilon_t\|^2$$

Negative diagnostics

More DDPM steps, simple low-pass action smoothing, and high-level RP1M fine-tuning did not improve the selected setting. These failures suggest piano contact timing is more important than generic smoothness or longer denoising at inference time.

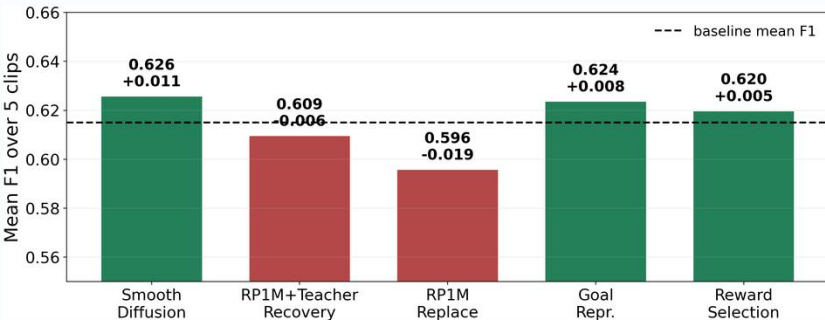
Results: Figures, Curves, and Tables

Single-task combined method



Onset reward + light smoothness + residual calibration improves Happy_8.

Generalist mean F1

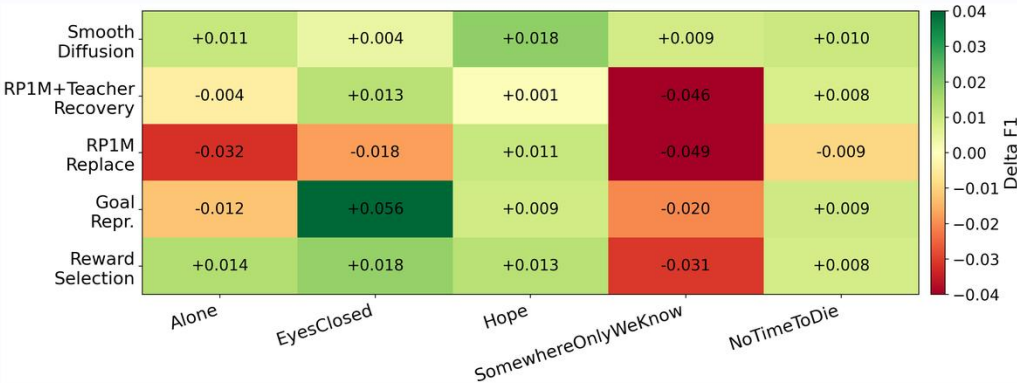


Smooth Diffusion Objective gives the best mean gain; Goal Representation is positive.

Numeric summary

Specialist method	F1	Delta	Outcome
combined specialist method	0.919	+0.046	final
Generalist method	F1	Delta	Result
Smooth Diffusion Objective	0.626	+0.011	best
RP1M Recovery + Teacher Reg.	0.609	-0.006	negative
RP1M Low-level Replacement	0.596	-0.019	negative
Goal Representation	0.624	+0.008	positive
Reward-based Rollout Selection	0.620	+0.005	positive
Broader generalist evidence			
Forester_1 RP1M linear-q: 0.6731 -> 0.6986			
12-song reward selection: 0.6095 -> 0.6165			
High-level RP1M / simple smoothing: negative diagnostics			

Per-clip generalist delta



Future Research

Stronger goal representation

- Encode onset timing, duration, local chord context, and finger-key hints.
- Give the generalist policy clearer future-note structure than raw goal states.

Contact-aware diffusion objective

- Add low-level losses for key-contact timing, press/release phase, and onset frames.
- Make denoised action chunks optimize piano contacts, not only trajectory similarity.

Possible objective

$$L = L_{\varepsilon} + \lambda_c (L_{\text{onset}} + L_{\text{contact}} + L_{\text{phase}})$$

Generalist residual correction

- Add a lightweight residual controller after the generalist diffusion action.
- Correct missed onsets, weak presses, and wrong-key errors while preserving coverage.